RED SEAL EXAM PREP

442A Industrial Electrician

Free 50-Question Mock Exam

Practice for the Red Seal / Certificate of Qualification exam

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https://redsealquiz.ca · Edition October 2026

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Mock Exam — answer all questions, then check the key

Q1. **What is a VFD (Variable Frequency Drive)?**

A) An electronic device that varies supply frequency and voltage

B) A device that varies the motor supply voltage to control speed

C) A device that controls motor torque only

D) A mechanical gear reducer for motor speed control

Q2. **Under CEC Section 18, which wiring method is acceptable in a Class I, Zone 1 (formerly Division 1) hazardous location?**

A) Threaded rigid metal conduit with flameproof fittings

B) Rigid PVC conduit with ordinary solvent-weld couplings

C) Electrical metallic tubing with set-screw couplings

D) NMD90 non-metallic sheathed cable stapled to framing

Q3. **What is an active harmonic filter?**

A) A passive LC trap tuned to shunt a single harmonic order, most often the fifth harmonic

B) A capacitor bank switched in only to correct displacement power factor at the main bus

C) An electronic unit that injects counter-harmonic current in real time to cancel harmonics

D) A series line reactor that limits harmonic current by adding impedance ahead of the drive

Q4. **A 442A electrician finds a 3-phase induction motor drawing Phase A=28A, Phase B=31A, Phase C=24A while running unloaded at rated voltage. What is the MOST likely cause?**

A) Voltage unbalance at the supply terminals

B) Worn motor bearings causing mechanical drag

C) Single-phasing from a blown control fuse

D) Motor is overloaded beyond nameplate rating

Q5. **What is the function of a contactor in a motor starter?**

A) To protect the motor from sustained overcurrent by opening the circuit on a thermal overload

B) To make and break the motor power circuit under load

C) To convert a single-phase supply into balanced three-phase power for the motor windings

D) To monitor motor speed and continuously adjust the applied voltage to hold that speed steady

Q6. **What is a timer instruction in PLC ladder logic?**

A) An instruction that restarts the processor after a set time

B) An instruction that changes a done bit at a preset time

C) A network module that synchronizes the clocks of several PLCs

D) A hardware clock module that sequences machine operations

Q7. **What is slip in an induction motor?**

A) The mechanical efficiency of the motor

B) The difference between rated and measured voltage

C) The difference between synchronous and rotor speed

D) The starting current relative to full-load current

Q8. **What is a delta-wye (Δ-Y) transformer connection and what is its advantage?**

A) Wye primary reduces short circuit current; Delta secondary increases voltage

B) The wye secondary provides a grounded neutral

C) Both delta and wye are identical in performance

D) Delta is for HV primary; Wye is only for motor loads

Q9. **What is a normally open (NO) contact in PLC ladder logic?**

A) A contact used only in emergency stop circuits

B) A contact that passes logic when its reference bit is TRUE

C) A contact that passes current only when its coil is de-energized

D) A physical contact that is open by default in the field

Q10. **What is a differential pressure (DP) transmitter used for?**

A) To measure absolute pressure in a vessel

B) To control the differential pressure in a boiler

C) To measure flow or level by sensing a pressure difference

D) To measure the voltage difference between two phases

Q11. **What is the difference between a "fail open" (FO) and "fail closed" (FC) control valve?**

A) FO opens at full stroke; FC closes at full stroke

B) FO is for liquid service; FC is for gas service

C) FO uses a butterfly valve; FC uses a globe valve

D) FO opens on loss of air/power; FC closes on loss

Q12. **An RTD (Pt100) temperature sensor reads -40°C in an oven that is clearly at room temperature (~22°C). A technician measures approximately 84 Ω at the RTD terminals at the DCS. What is the MOST likely fault?**

A) The DCS analog card has failed

B) RTD element has failed open

C) RTD calibration drift from overheating

D) RTD extension wires are partially shorted

Q13. **An electrician is de-energizing a 600 V motor control centre to make it electrically safe before maintenance. Under the 2024 edition of CSA Z462, where must the absence of voltage be verified?**

A) Only at the main disconnect feeding the MCC

B) At one test point chosen on each feeder

C) Only on the line side of each lockout device

D) At each point of work on the equipment

Q14. **Switchgear feeding a large pump motor has been replaced. A phase rotation tester at the motor starter reads the same A-B-C sequence as before the work. Why must the motor still be bumped and the shaft direction observed before the pump goes back into service?**

A) Because the tester reads sequence only above the motor's rated voltage

B) Because the supply sequence alone does not fix which way the shaft turns

C) Because bumping is the only way to confirm the overload is sized right

D) Because a rotation tester cannot read the sequence on a loaded circuit

Q15. **What is the purpose of a power factor correction capacitor bank?**

A) To regulate voltage at the service entrance

B) To filter voltage harmonics from VFDs

C) To supply reactive power locally

D) To store energy for emergency power

Q16. **What is a motor's breakdown torque?**

A) The maximum torque the motor can develop at rated voltage

B) The torque the motor produces at zero speed in the instant the contactor first closes

C) The torque loading at which the stator windings overheat and burn out under sustained overload

D) The torque delivered continuously at nameplate full load speed and rated nameplate current

Q17. **Before touching conductors that have been isolated and locked out, a worker tests for absence of voltage. Why must the tester be proved on a known live source both before and after that test?**

A) To prove that the tester itself was working when it read zero

B) To confirm the tester is set to the correct voltage range first

C) To satisfy the requirement that two workers witness the test

D) To discharge any stored charge left in the tester's own leads

Q18. **What is a control valve Cv (flow coefficient)?**

A) The valve stroke in percentage per volt of signal

B) The valve's closing velocity in m/s

C) The valve's flow capacity at 1 psi pressure drop

D) The calibration verification number of the valve

Q19. **What is the HART protocol and what is its primary advantage?**

A) Hardwired Addressable Relay Technology — a relay-based hardwired interlock scheme

B) High-speed Analog Real-Time Transmission — faster than a standard 4-20 mA loop

C) A fibre-optic link that replaces the 4-20 mA loop between field and control room

D) Highway Addressable Remote Transducer — digital data on the same 4-20 mA loop

Q20. **What is OPC-UA in industrial automation?**

A) OPC Unified Architecture — an open data exchange standard

B) A proprietary programming language that runs only on Siemens S7 series controllers

C) Optimal Power Control — Utility Automation, a plant demand management scheme

D) Override Process Control — Universal Application, a manual DCS override mode

Q21. **What is the difference between AC and DC electricity?**

A) AC reverses direction periodically; DC flows one way

B) AC is only for power transmission; DC is for motors

C) DC is safer than AC for all applications

D) AC always has higher voltage than DC

Q22. **What personal protective equipment is required for energized work on 600 V equipment?**

A) No PPE is needed at 600 V, which is low voltage

B) Leather work gloves and a hard hat are enough

C) Safety glasses and CSA-approved work boots only

D) Arc-rated PPE with insulated gloves and tools

Q23. **A delta-delta bank of three identical 50 kVA single-phase transformers supplies a plant. One transformer fails and is disconnected, and the bank is reconnected open delta on the two remaining units. What is the largest balanced three-phase load, in kVA, that the open-delta bank can carry without overloading either transformer?**

A) 129.9 kVA

B) 75.0 kVA

C) 86.6 kVA

D) 100.0 kVA

Q24. **A wound-rotor induction motor is connected with external resistance in the rotor circuit. After the resistance is short-circuited, motor RPM increases from 1,680 to 1,740 under the same load. What does this confirm?**

A) The motor had a shorted stator winding

B) Synchronous speed of this motor is 1,740 RPM

C) External resistance was increasing slip and reducing speed

D) The motor was single-phasing before the resistance change

Q25. **What is motor efficiency and what does IE3 mean?**

A) IE3 = 3-phase efficiency; efficiency refers to power factor only

B) IE3 = Premium Efficiency; efficiency = mechanical output / electrical input

C) IE3 = third generation motor; efficiency is ratio of mechanical output to electrical input

D) IE3 = insulation class; efficiency is ratio of torque to speed

Q26. **What is the purpose of a distribution transformer in an industrial facility?**

A) To step down supply voltage to utilization voltage

B) To filter the harmonic currents produced by the plant's variable frequency drives

C) To correct the plant's overall power factor and reduce the utility demand charge

D) To generate electrical power on site to carry the plant through a utility outage

Q27. **What is PLC scan time and why does it matter in safety-critical applications?**

A) The time between scheduled PLC maintenance — longer scan = more wear

B) The time for the PLC to boot after power loss

C) The time for one complete scan cycle — limits response speed

D) The time for the PLC to communicate with the HMI

Q28. **Under the 2024 edition of CSA Z462, when may energized work go ahead instead of first putting the equipment into an electrically safe work condition?**

A) Only where doing so is not practicable

B) Only where it is rated at 600 V or lower

C) Only where arc-rated clothing is being worn

D) Only where a qualified person is watching

Q29. **What does FLA or FLC stand for on a motor nameplate?**

A) Full Line Ampacity

B) Frequency Limit Adjustment

C) Full Load Amperes

D) Fuse Load Allowance

Q30. **What is the formula for three-phase power?**

A) $P = V \times I$ (same as single phase)

B) $P = \sqrt{3} \times V_L \times I_L \times PF$ (line values)

C) $P = V_L \times I_L / \sqrt{3}$

D) $P = 3 \times V \times I \times PF$ (where V = line voltage)

Q31. **A 600 V switchboard has a nameplate rating below 1200 A and no draw-out parts. What minimum working space does the CEC require in front of it?**

A) 600 mm (24 in)

B) 1.0 m (39 in)

C) 1.5 m (60 in)

D) 900 mm (36 in)

Q32. **What is a short circuit current rating (SCCR) and why is it important for industrial panels?**

A) The minimum current needed for the circuit to operate

B) The overcurrent protection setting for the main breaker

C) The panel's maximum continuous load current rating

D) The maximum fault current a panel can safely withstand

Q33. **A technician uses a clamp meter to measure current on a 3-phase motor with star (Y) connection. Each phase measures 18A. What is the line current feeding this motor?**

A) 54A (18 × 3)

B) 18A (line = phase)

C) 31.2A (18 × √3)

D) 10.4A (18 / √3)

Q34. **What is a ground fault circuit interrupter (GFCI) and where is it required?**

A) A device that protects branch circuit conductors from sustained overload and thermal damage

B) A device that limits the arc-flash incident energy released at the panel during a fault

C) A fuse element that clears bolted short circuits and high-current faults only

D) A device that opens the circuit on a ground-fault imbalance of about 6 mA

Q35. **A technician is troubleshooting a PLC-controlled system. The physical motor runs but the PLC output LED is OFF. The motor starter coil is confirmed energized from an external source. What does this indicate?**

A) The processor is in run mode but the scan is halted

B) The status LED on the output module is defective

C) The motor is fed through a bypass, not by the PLC

D) The output module has failed and must be replaced

Q36. **A 442A electrician measures a single-phase 120V AC circuit drawing 15A with a power factor of 0.75 lagging. What is the true (real) power consumed?**

A) 1,800 W

B) 1,190 W

C) 2,400 W

D) 1,350 W

Q37. **What is the purpose of a safety relay or safety PLC?**

A) To supply backup control power so the machine keeps running when the main PLC loses supply

B) To protect the PLC input and output cards from voltage spikes and inductive switching transients

C) To automatically reset and restart the machine as soon as the emergency stop button is pulled out

D) To monitor safety functions and force a safe state on fault

Q38. **What is the difference between a TEFC and ODP motor enclosure?**

A) TEFC is for DC motors; ODP is for AC motors

B) TEFC is explosion-proof; ODP is for outdoor use

C) TEFC is sealed and fan cooled; ODP has ventilation openings

D) They are interchangeable — enclosure type has no practical significance

Q39. **What is an equipotential bonding mat used for in industrial work?**

A) To protect workers from step and touch potential

B) To bond the conduit and raceway system back to the driven ground electrode at the service entrance

C) To measure the resistance of bonding conductors and the ground grid during commissioning tests

D) To provide a clean insulated surface for storing rubber gloves and live-line tools between tasks

Q40. **What is the role of a motor nameplate?**

A) To log the running hours and service history of the motor

B) To give the rated data used for installation and protection

C) To show the maker's brand and catalogue number, nothing more

D) To record the warranty period and the serial number for claims

Q41. **A Type J thermocouple is installed in a furnace. The DCS reads about 305°C while a calibrated RTD reference beside it shows 230°C. The error is high, repeatable, and grows as the furnace heats up. What is the MOST likely cause?**

A) Ambient temperature at the terminal block is high and is being added on top of the process reading

B) The thermocouple is installed with reversed polarity, which doubles the millivolt signal reaching the controller

C) The thermocouple extension wire has an open break in the run, which adds a fixed error to the reading

D) The controller is configured for Type K instead of Type J thermocouple

Q42. **What is a 2-wire vs 4-wire transmitter?**

A) 2-wire transmitters are for digital signals; 4-wire for analog

B) 2-wire is loop-powered; 4-wire has a separate power supply

C) Wire count refers to the number of calibration points only

D) 2-wire = only positive terminal; 4-wire = both terminals

Q43. **What happens if a current transformer (CT) secondary is open-circuited while the primary is energized?**

A) The primary current collapses toward zero because the transformer's magnetic circuit is broken

B) Dangerously high voltage appears on the secondary

C) The loss of secondary current is detected and the protective relay trips the feeder offline

D) Nothing hazardous happens — the CT simply stops producing an output until it is reconnected

Q44. **A technician performs a voltage drop test on a 120V, 20A branch circuit supplying a load 50m from the panel. The voltage drop measured is 8V (6.7%). What corrective action should be taken per CEC requirements?**

A) Add a voltage regulator at the load end

B) Replace the load with a lower-power model

C) No action needed — CEC allows 10% drop

D) Increase the conductor size for the run

Q45. **What is impedance (Z) in AC circuits?**

A) The maximum instantaneous current an AC circuit can carry when it is driven at resonance

B) The phase angle in degrees by which the current waveform lags or leads the applied voltage

C) The purely resistive component of the circuit, which an ohmmeter reads directly across the load

D) The total opposition to current flow in an AC circuit

Q46. **What is the purpose of a star-delta (Y-Δ) starter?**

A) To step up voltage for high-voltage motor starting

B) To reduce starting current by starting in star, then switching to delta

C) To allow the motor to run in both clockwise and counterclockwise directions

D) To vary motor speed by changing the number of poles

Q47. **What is the purpose of a motor's thermal protection (thermistor or thermostat)?**

A) To measure motor frame vibration

B) To measure motor shaft speed

C) To protect against phase loss only

D) To directly sense winding temperature

Q48. **A 4-20mA loop-powered transmitter reads 3.6mA with the process at 0% (minimum range). What does this indicate and what should the technician do FIRST?**

A) Fault signal — check its diagnostics, supply and sensor

B) Broken signal wire — replace immediately

C) Normal — some transmitters operate below 4mA at process zero

D) PLC analog input card has failed

Q49. **What is Ohm's Law?**

A) $V = I \times R$

B) $P = I \times R$

C) $R = V \times I$

D) $I = V \times R$

Q50. **What is an E-stop (emergency stop) circuit and why must it be hardwired?**

A) A hardwired circuit that removes power regardless of PLC state

B) A normally open contact wired in parallel with the start button

C) A software command in the PLC program that stops the motors

D) A wireless emergency stop button linked to the PLC by radio

Answer Key & Explanations

> Scoring: 35 correct out of 50 is 70%. These 50 questions come from the 130-question 442A bank; the last page shows what the complete printable bank contains.

Q1. Answer: A

A VFD converts fixed AC to variable frequency and voltage AC, allowing precise motor speed control. Speed is proportional to frequency (n = 120f/P). Also called variable speed drive (VSD) or adjustable frequency drive (AFD).

Q2. Answer: A

A Zone 1 location can hold an explosive gas atmosphere during normal operation, so the wiring must neither become a source of ignition nor let an ignition inside the system propagate out to the surrounding atmosphere. Threaded rigid metal conduit with flameproof (explosion-proof) fittings does both: the long threaded joints act as flame paths that cool escaping gases below the ignition temperature of the atmosphere outside. The other Zone 1 method used every day in Canadian plants is hazardous-location certified cable, such as HL-marked Teck 90 terminated in certified cable glands. Set-screw tubing, solvent-welded PVC and non-metallic sheathed cable offer no flame path and no certified gland, so none of them is acceptable.

Q3. Answer: C

VFDs and other non-linear loads draw current in pulses rather than sinusoidally - a typical 6-pulse drive draws current only near the peaks of the sine wave. On a three-phase drive the harmonics left are mainly the 5th and 7th: Rockwell notes that balanced three-phase rectifier loads such as an AC drive produce no third harmonic and no harmonic with 3 as a multiple (3rd, 9th, 15th and so on) - triplen harmonics, which are not present in most AC drives - so a balanced three-phase drive does not put triplen harmonic current on the system. Harmonic current still costs the installation: it adds I^2R loss in the wiring and higher flux in transformer iron, and through the source impedance it distorts the bus voltage. An active harmonic filter uses current sensors and a switching circuit much like a drive's inverter to add the sine wave complement of the current it measures, making the current upstream look sinusoidal. Passive mitigation — line reactors, DC link chokes and passive filters — does reduce harmonics, but with fixed inductors and capacitors rather than by measuring the current and correcting it.

Q4. Answer: A

A small voltage unbalance shows up as a much larger current unbalance. Current unbalance runs roughly 6 to 10 times the voltage unbalance, on a percent basis. These three readings average 27.7A and the worst phase (the 24A phase) is about 3.7A off that average, so the current unbalance is about 13% - which points back to a voltage unbalance of only about 1 to 2% at the supply, small enough to be missed unless the line voltages are actually measured. Worn bearings cause noise and drag but do not unbalance the phase currents. Single-phasing would show one phase near zero. A genuine overload raises all three phases together, and this motor is running unloaded.

Q5. Answer: B

A contactor is a heavy-duty, magnetically operated switch with large contacts rated to make and break load current repeatedly. It is the power-switching component of a motor starter.

Q6. Answer: B

Timer instructions count elapsed time against a preset value, and each type drives its done bit differently. A TON, timer on-delay, sets its done bit once the accumulated time reaches the preset while the rung stays true. A TOF, timer off-delay, sets its done bit as soon as the rung goes true and resets it once the accumulated time reaches the preset after the rung goes false - nothing is set at the preset. An RTO, retentive timer, keeps its accumulated time when the rung goes false and must be cleared with a RES instruction. Timers are used for timed sequences, delay functions and time-based control.

Q7. Answer: C

Slip is expressed as a percentage: Slip = (Ns - Nr) / Ns × 100%. At no load, slip is small. At full load, typical slip is 2–5%. Slip allows the rotor conductors to experience a changing magnetic field, inducing the rotor current that creates torque.

Q8. Answer: B

A delta-wye transformer has the primary wound in delta, with no neutral, and the secondary wound in wye with a neutral point that is grounded. That gives a 4-wire secondary carrying both the three-phase line-to-line voltage (for example 600 V) and a line-to-neutral voltage (347 V, since 600/√3 = 346.4), so single-phase loads and the system ground reference are taken from the same transformer. Wrong answers: a wye primary does not reduce short-circuit current and a delta secondary does not raise voltage — available fault current is set by the supply impedance and by the transformer's impedance and turns ratio, not by the winding configuration; a wye secondary is not restricted to motor loads, since its whole value is that it also supplies line-to-neutral single-phase loads; and the two connections are not identical in performance, because only the wye secondary offers a neutral. For contrast, a wye-delta bank places the neutral on the primary side only and leaves a three-wire secondary with no neutral, while a delta-delta bank has no neutral on either side. The delta winding also gives triplen harmonic currents a path to circulate, keeping them out of the other winding. 600 V/347 V systems in Canada use delta-wye transformers so that 347 V single-phase lighting branch circuits can be supplied from the same transformer as 600 V three-phase motor loads.

Q9. Answer: B

In ladder logic, a NO contact (XIC instruction in Allen-Bradley) allows rung continuity when the corresponding bit in memory is 1 (TRUE). It represents the state when the associated coil or input is energized.

Q10. Answer: C

DP transmitters measure the difference between two pressure taps. With an orifice plate restriction, DP indicates flow (square root relationship). For level measurement, DP corresponds to liquid column height × density.

Q11. Answer: D

Fail-safe position is determined by process safety analysis. Fail open (FO): the spring pushes the valve open on loss of instrument air (e.g., cooling water valves must stay open on failure). Fail closed (FC): the spring closes the valve (e.g., fuel gas valves must close on failure).

Q12. Answer: D

A shorted RTD lead appears as lower resistance than actual temperature. A short circuit somewhere in the 3-wire RTD extension leads reduces measured resistance below actual, making the DCS calculate a lower (incorrect) temperature. Pt100 at 22°C ≈ 108.6 Ω, but at −40°C it is 84.27 Ω (IEC 60751) — the ~84 Ω measured means a partial short in the leads is bypassing ~25 Ω of the element's true resistance, so the DCS computes −40°C.

Q13. Answer: D

At each point of work. CSA Group's overview of the 2024 edition says the requirement to verify the absence of voltage must now be met 'at each point of work' (Clause 4.2.5). A reading at the main disconnect or on the line side of a lockout device proves the state of that spot only, and one test point per feeder leaves the other points of work unproved. Verification is the last proof, not a substitute for the steps before it: Ontario's Construction Projects regulation also requires the power supply to be disconnected, locked out of service and tagged, and hazardous stored electrical energy to be adequately discharged or contained, before the work begins, and requires the worker to verify both before beginning (O. Reg. 213/91, s. 190(4), (5) and (7)).

Q14. Answer: B

A sequence test proves what the supply is doing, not what the shaft will do. A phase rotation tester tells you the order in which the three supply phases reach their peaks at the point where you clipped on. Confirming that after switchgear work is worth doing, because a reversed supply sequence would reverse every motor on the bus at once. But it says nothing about which way one particular shaft turns, and that depends on three more things the tester never sees: how the motor leads were landed in the terminal box, how the winding was connected internally if the motor has been rewound, and whether a coupling or gearbox between motor and pump reverses direction. Only watching the shaft during a momentary bump settles it. The stakes are that a centrifugal pump run backwards still moves some liquid and can look plausible on a flow indicator, so the error survives long enough to do damage. Rotation testers work at normal system voltage and do not need the circuit unloaded, and overload sizing comes from the nameplate full-load current, not from a bump.

Q15. Answer: C

Inductive loads (motors, transformers) draw reactive power (kVAR) from the utility. Capacitor banks supply this reactive power locally, reducing the reactive current drawn through supply cables, improving power factor, and reducing utility demand charges and penalties.

Q16. Answer: A

Breakdown torque (pull-out torque) is the maximum torque an induction motor can develop at rated voltage before speed drops into an unstable region. If load exceeds this value, the motor stalls. Typically 200–300% of full-load torque.

Q17. Answer: A

A voltage tester that has failed reads zero on a live conductor exactly as it reads zero on a dead one. A blown fuse, a broken lead, a flat battery, a damaged range switch - none of them announce themselves, and every one of them turns the instrument into a device that says 'safe' about everything. Proving the tester on a known live source immediately before the absence-of-voltage test, taking the test, then proving it again on the known live source afterwards is what establishes that the zero reading meant the conductors were dead rather than that the instrument was dead. The second proving matters as much as the first, because an instrument can fail during the test. This live-dead-live check is a step in establishing an electrically safe work condition, after the disconnecting devices have been opened, locked out and visually verified. Selecting the right range matters but detects nothing about a failed instrument, nothing needs discharging from the leads, and a second person watching does not verify the meter.

Q18. Answer: C

Cv is the flow capacity of a valve — the volume of water (US gallons per minute) that flows through it with a 1 psi pressure drop. Sizing: $Cv = Q \times \sqrt{(SG/\Delta P)}$ where Q is flow in gpm, SG is specific gravity, ΔP is pressure drop in psi. Larger Cv = larger flow capacity.

Q19. Answer: D

HART (Highway Addressable Remote Transducer) superimposes a digital frequency-shift-keyed signal — roughly ±0.5 mA at the 1200 Hz and 2200 Hz Bell 202 tones — on top of the 4–20 mA loop current. Because the modulation averages to zero, the analog measurement keeps controlling the process while a handheld communicator or an asset-management system reads and writes configuration, runs diagnostics and pulls secondary variables over the same two wires. That is the advantage: digital access with no additional wiring and no disturbance to the analog control signal. Wrong answers: HART is not a faster substitute for the analog loop — the digital channel is deliberately slow, which is why it is used for configuration and diagnostics rather than for closed-loop control; it is not a fibre-optic control-room network, and it does not replace the 4–20 mA loop, it rides on it; and the acronym does not stand for a relay interlock scheme.

Q20. Answer: A

OPC-UA (OPC Unified Architecture) is an open, platform-independent, secure data exchange standard for industrial automation. It enables interoperability between different vendor systems, PLCs, SCADA, and cloud platforms.

Q21. Answer: A

AC (alternating current) reverses polarity/direction at the supply frequency (60 Hz in North America = 120 reversals/second). DC (direct current) flows in one constant direction only. AC is used for power distribution; DC for electronics, batteries, and VFD internal circuits.

Q22. Answer: D

Energized work on a 600 V industrial system carries both a shock hazard and an arc flash hazard, the two categories into which the 2024 edition of CSA Z462 sorts electrical hazards (CSA Group). The arc-rated clothing and face protection are chosen by one of two methods, and the two are not mixed on the same equipment: an incident energy analysis, or the standard's arc flash PPE tables (the 2015 edition's rule; the 2024 edition calls the second route the arc flash PPE selection table method). Either route ends in the same kind of kit - arc-rated clothing and face protection, rubber insulating gloves with protectors, and insulated tools, which the 2015 edition requires inside the restricted approach boundary. Ontario's Construction Projects regulation adds that tools, devices and equipment, including PPE, used on or near energized exposed parts must be designed, tested, maintained and used so as to provide adequate protection (O. Reg. 213/91, s. 192). Energized work also needs an energized electrical work permit unless the activity is on the 2024 edition's list of exemptions (Clause 4.3.2.3).

Q23. Answer: C

An open-delta (V) bank is rated at the square root of 3 times the kVA of one unit: 1.732 x 50 = 86.6 kVA. In the V connection each remaining winding sits directly in series with a line, so it carries the full line current at the full line voltage. With each winding held to its 50 kVA rating, the three-phase output is the square root of 3 x line voltage x line current = 1.732 x 50 = 86.6 kVA, against 3 x 50 = 150 kVA for the closed bank - the bank keeps 57.7% (1 divided by the square root of 3) of its original rating. At that load each unit carries rated current at rated voltage, so each is at 100% of its own nameplate, yet the two units together yield only 86.6 kVA of their combined 100 kVA nameplate: the current in each winding is displaced 30 degrees from that winding's voltage, and cos 30 degrees = 0.866, so the two kVA ratings do not add arithmetically. 100 kVA is therefore the survivors' nameplate sum, which would put 57.7 kVA on each 50 kVA winding (115%). 129.9 kVA applies the 86.6% figure to the wrong base - the original 150 kVA bank instead of the two survivors. 75 kVA comes from stacking both derating factors (150 x 0.577 x 0.866), or from assuming the loss simply halves the bank. If the original 150 kVA load is left connected, each unit carries about 173% of rated current (150 divided by 86.6), which is why open delta is an emergency measure: cut the load to 86.6 kVA until the third unit is replaced.

Q24. Answer: C

External rotor resistance increases slip, reducing speed. Wound-rotor motors use external resistance to limit starting current and control speed. When resistance is removed (short-circuited) for normal operation, slip decreases and speed approaches synchronous speed. 1,800 RPM synchronous at 60 Hz for a 4-pole motor; 1,740 RPM represents normal running slip.

Q25. Answer: B

Motor efficiency = (output HP × 746) / (input watts) × 100%. IEC 60034-30 defines efficiency classes: IE1 (standard), IE2 (high), IE3 (premium), IE4 (super premium). IE3 is required for most motors in Canada under energy efficiency regulations.

Q26. Answer: A

Distribution transformers step down the high-voltage utility supply (e.g., 13.8 kV) to the utilization voltages used by plant equipment. Common industrial: 13.8 kV → 600V for large motors/feeders, 600V → 120/208V for lighting and controls.

Q27. Answer: C

Scan time — the time the controller takes to complete one scan: input scan, program scan, output scan and housekeeping — sets how quickly the PLC and its control loop can respond to process changes, and the full input-to-output response (throughput) adds input and output delays on top of it. In safety applications, if the response is too slow, a hazardous condition may persist for longer than acceptable. A safety controller bounds it: Rockwell's GuardLogix manual says the safety-task reaction time is at most the safety task period plus the safety task watchdog, and a safety task that overruns its watchdog faults the controller and turns the outputs off.

Q28. Answer: A

Only where establishing that condition is not practicable. CSA Group's overview of the 2024 edition says the exception to the requirement to establish an electrically safe work condition (Clause 4.1.6.3) was revised to a single exception - when it is 'not practicable' to do so - to align with Canadian occupational health and safety regulations, and that the requirements for the condition were reorganized into Clauses 4.1.6.1 to 4.1.6.3. A voltage rating, a set of arc-rated clothing or a qualified person watching is not that exception: arc-rated clothing is protection for work that is still treated as energized, not a reason to skip the safe condition. Where energized work does go ahead, the 2024 edition still calls for an energized electrical work permit unless the activity is on its list of exemptions (Clause 4.3.2.3).

Q29. Answer: C

FLA (Full Load Amperes) or FLC (Full Load Current) is the motor's rated current drawn at nameplate voltage and full mechanical load. Used for sizing conductors, overload protection, and motor starter components.

Q30. Answer: B

Three-phase real power: $P = \sqrt{3} \times V_L \times I_L \times \cos\theta$. Where V_L = line-to-line voltage, I_L = line current, $\cos\theta$ = power factor. This applies to balanced 3-phase systems.

Q31. Answer: B

Rule 2-308(1) calls for a minimum working space of 1 m with secure footing in front of electrical equipment such as switchboards, panelboards, control panels and motor control centres, and that space is kept clear of obstructions. The step up to 1.5 m comes from Rule 2-310(2), and it follows the equipment nameplate rating rather than the overcurrent setting: it applies where the board is rated 1200 A or more, or over 750 V, and a person inside the electrical room or the space around the equipment could not leave without passing a potential failure point on the path to the exit. A 600 V board rated below 1200 A meets neither of those ratings, so the 1 m figure stands. Equipment that has draw-out parts adds the depth of those parts on top of whichever figure applies, which is why the question rules them out. The trap option 900 mm (36 in) is the working-clearance figure from US NEC 110.26, whereas the Canadian minimum in front of this switchboard, set by Rule 2-308(1), is 1 m (39 in).

Q32. Answer: D

SCCR is the maximum prospective fault current an electrical assembly — a panel, MCC or drive — can withstand without damage beyond acceptable limits. It is a withstand rating belonging to the assembly and its non-protective components, and it is distinct from the interrupting rating (AIC) of the fuses or breakers inside it, which is the fault current those protective devices can themselves safely clear. Both matter and they are specified separately: the assembly has to be rated for at least the available fault current at the point of installation, and the overcurrent device has to be able to interrupt it. An under-rated assembly can rupture busbars and blow open its enclosure instead of letting the fault be cleared cleanly.

Q33. Answer: B

In a star (Y) connected motor, line current equals phase current. In Y connection: I_line = I_phase = 18A. Only voltage differs: V_line = V_phase × √3. This is opposite to delta: in delta, I_line = I_phase × √3, but V_line = V_phase. The clamp meter on each supply line will read 18A.

Q34. Answer: D

A GFCI compares the current leaving on the ungrounded conductor with the current returning on the neutral; any difference is going to ground, possibly through a person. A Class A device must interrupt the circuit when that imbalance is 6 mA or more, and must not interrupt it when the imbalance is 4 mA or less. That is why a GFCI protects people while a fuse or breaker, sized to protect the conductor, will never open at milliamp levels. The threshold and the trip time are two separate specifications: the maximum permitted trip time is an inverse curve, several seconds at the 6 mA threshold and only about 20 ms once the fault current passes a few hundred milliamps. In Ontario, Rule 26-704 requires Class A GFCI protection for 5-15R and 5-20R receptacles installed outdoors within 2.5 m of finished grade, including block-heater receptacles in parking lots.

Q35. Answer: C

If the motor is running while the PLC output LED is off, and the starter coil has been confirmed energized from an external source, then something other than the PLC output is holding that coil in. A bypass switch, a hand-off-auto selector left in hand, or a parallel wire can energize the starter directly. This is a safety concern: if the PLC output is supposed to be the only control of that motor, the alternate path has to be found, documented, and either removed or interlocked. Trace the starter coil wiring back to its source. The confirmed external source is what rules out the two module faults offered here - a shorted output device or a dead status light would both put the source back inside the module.

Q36. Answer: D

Real power = V × I × PF = 120 × 15 × 0.75 = 1,350 W. Apparent power is what the conductors actually carry: V × I = 1,800 VA, and that figure is in volt-amperes, not watts. Dividing by the power factor instead of multiplying gives 2,400 W, a number larger than the apparent power, which is impossible. Using sin θ in place of cos θ returns about 1,190, but that is the reactive power in VAR, not the real power in watts. Only the real power does useful work; the reactive component is stored and returned each cycle by the circuit inductance.

Q37. Answer: D

Safety relays/PLCs (designed and rated to IEC 62061 or ISO 13849-1) monitor safety functions such as E-stops, light curtains, and guard interlocks using redundant, self-diagnostic circuits. On fault detection, they cut power to hazardous functions so the machine reaches a safe state before harm occurs. **In Canada those standards reach the shop floor through CSA Z432, Safeguarding of Machinery**, which requires safety-related parts of control systems on newly manufactured machinery to provide functional safety performance as determined by ISO 13849-1 or IEC 62061.

Q38. Answer: C

TEFC (Totally Enclosed Fan Cooled) motors are sealed against contaminants and have an external cooling fan — suitable for dusty, wet, or contaminated environments. ODP (Open Drip-Proof) motors are ventilated (cooling air passes through) — suitable for clean, indoor, protected locations only.

Q39. Answer: A

Equipotential bonding mats ensure that the floor, equipment, and person are at the same electrical potential, preventing dangerous current flow through the worker during a ground fault. They are commonly used on metal grating and in substations, where step and touch potential hazards are greatest.

Q40. Answer: B

The nameplate carries the motor's rated data, and installation and protection are worked from those ratings. NEMA MG-1 requires certain information on the nameplate of single- and three-phase motors, and the nameplate data can include horsepower, volts, amps, hertz, phase, rpm, insulation class, enclosure, frame size, efficiency, service factor, power factor, duty, ambient, code and design (SkilledTradesBC apprenticeship text on BCcampus). The nameplate full-load current is the figure the overload heater is selected from (BCcampus Basic Motor Control), and under CEC Rule 28-106 1) the branch-circuit conductors of a motor on continuous duty must have an ampacity of at least 125% of its full-load current rating (CMEQ installation bulletin BTI-015, 2019; Alberta STANDATA 24-ECB-028, on the 2024 Code, advises caution where Table 27 brings conductors rated below 125% of the nameplate current into consideration). The maker's name and catalogue number are on it too, but identification is not its whole job, and a nameplate is a fixed plate of ratings, not a record that is kept up to date.

Q41. Answer: D

The wrong thermocouple type in the controller produces a systematic error that grows with temperature. Type J (iron/constantan) puts out more millivolts per degree than Type K (nickel-chromium/nickel-alumel). At 230°C a Type J junction produces roughly 12.4 mV, but a controller looking that value up in a Type K table — where about 12.4 mV corresponds to a little over 300°C — reports a much higher temperature. Because the two curves diverge as temperature climbs, the error grows with the process, which is exactly what the technician sees. The other explanations do not fit the evidence: a cold-junction or terminal-block problem shifts the reading by a roughly constant amount and does not scale with process temperature, an open extension wire drives the input to an upscale or downscale burnout alarm rather than a plausible number, and reversed polarity makes the reading move the wrong way as the furnace heats, not read high.

Q42. Answer: B

A 2-wire (loop-powered) transmitter draws its operating power from the 4–20 mA loop current. A 4-wire transmitter has a separate power supply (120V AC or 24V DC) and outputs a 4–20 mA signal independently.

Q43. Answer: B

With the secondary open, the primary current acts solely as magnetizing current, fully magnetizing the CT core. The core saturates and the collapsing magnetic flux induces extremely high, potentially lethal voltage spikes (thousands of volts) on the open secondary terminals.

Q44. Answer: D

CEC Rule 8-102 requires that voltage drop not exceed 3% in a feeder or branch circuit, and not exceed 5% overall from the supply side of the consumer's service to the point of utilization. This is mandatory 'shall not exceed' language, not a design suggestion, and an inspector can write it up. At 6.7% the branch circuit is over the limit, so the conductor is undersized for a 50 m run at 20 A. Voltage drop is current times resistance, and the only listed action that reduces the resistance of the run is a larger conductor: more cross-sectional area means less resistance per metre and less drop. A regulator at the load end hides the symptom while leaving the same heating in the conductor, and changing the load does not correct an installation that fails the rule.

Q45. Answer: D

Impedance $Z = \sqrt{(R^2 + (XL\text{-}Xc)^2)}$. It is the complete opposition to AC current — the vector sum of resistance (R), inductive reactance (XL), and capacitive reactance (Xc) at a specific frequency.

Q46. Answer: B

Y-Δ starting connects windings in star (Y) first, applying $1/\sqrt{3}$ of line voltage per winding, reducing starting current and torque to 1/3 of DOL values. After acceleration, the starter switches to delta (Δ) for full power.

Q47. Answer: D

PTC thermistors or bimetallic thermostats embedded in motor windings directly measure winding temperature. When temperature exceeds the setpoint, they trip or signal the motor control circuit to stop the motor before insulation damage occurs.

Q48. Answer: A

4mA is the live zero - a steady 3.6mA is below the measuring range and sits in the band NAMUR NE 43 reserves for failure signals. In a 4-20mA system, 4mA = 0% process. Transmitter makers that follow NE 43 set out the levels: Endress+Hauser's TMT82, for example, lets the signal fall only to 3.8mA when the process goes below range (and rise only to 20.5mA above it), and drives it to 3.6mA or less, or 21mA or more, to signal a failure such as a sensor fault - Pyromation's transmitter uses the same NE 43 levels. So some transmitters do go below 4mA, but only to 3.8mA and only when the process is below the bottom of its range; a transmitter holding 3.6mA with the process at 0% is reporting a fault, not a reading. Input cards flag the same region: Rockwell's HART analog input module, for one, sets an underrange flag at 3.6mA or less and an overrange flag at 21.0mA or more. A broken signal wire opens the loop and drops the current to about 0mA, not 3.6mA. So the first move is to find out what the transmitter is reporting: read its diagnostics (Endress+Hauser's standard diagnostic functions cover cable open-circuit and short-circuit of the sensor wires, incorrect wiring and internal device errors), and check the supply at its terminals - E+H's low-voltage detection drops the output below 3.6mA when the supply voltage falls below the required value. Clear the cause; a zero trim or recalibration comes only after the fault signal has gone, because 3.6mA here is a failure signal, not a measurement to be trimmed.

Q49. Answer: A

Ohm's Law: V = I × R. Voltage (volts) = Current (amperes) × Resistance (ohms). Derivations: I = V/R, R = V/I. Applies to DC and resistive AC circuits.

Q50. Answer: A

An E-stop must be a hardwired circuit built on normally-closed, direct-opening contacts, because a stop that exists only in software fails with the processor or the program that hosts it. The contacts are wired in series in the control circuit so that a broken conductor or a loose terminal drops the circuit out and stops the machine, whereas a normally open contact would leave the same fault silently disabling the stop. Machine stopping functions are classified as stop category 0, an immediate removal of power, or stop category 1, a controlled stop followed by removal of power. A hardwired E-stop interrupts the motor control circuit and removes hazardous energy independently of what the PLC program is doing.

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